1 Fig. 1.1 shows a force–extension graph for a spring.

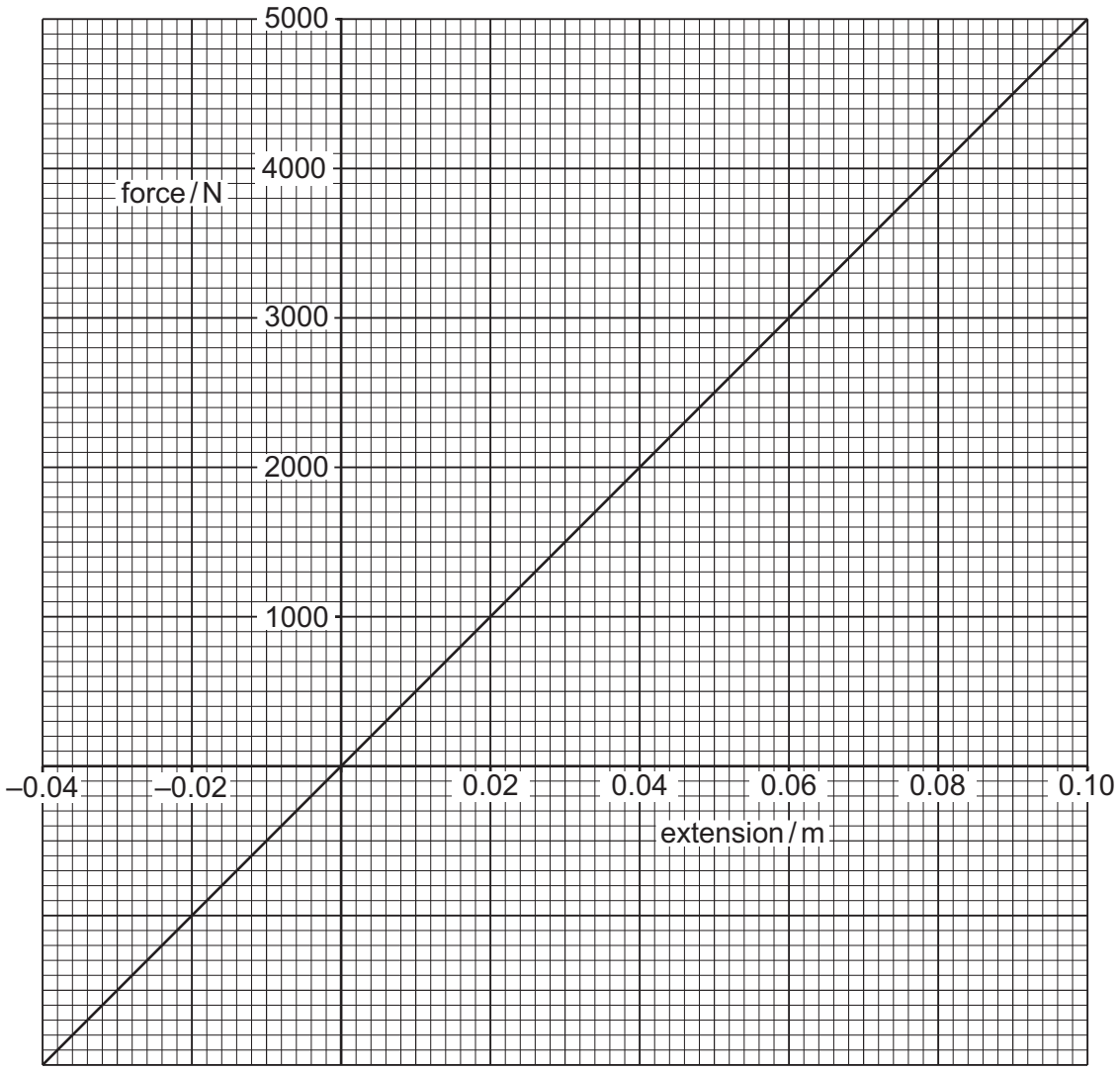

Fig. 1.1

(a) Calculate the spring constant k of the spring.

k = ... [2]

(b) A student states that the spring has **not** reached the limit of proportionality when a force of 4500 N is applied to it.

State how the graph shows that this statement is true.

.....

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(c) Springs can be compressed by forces. The spring described by Fig. 1.1 is compressed by a force F and has an extension of –0.025 m.

Determine F.

F = ... [2]

(d) State whether force is a scalar quantity or a vector quantity. Explain your answer.

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